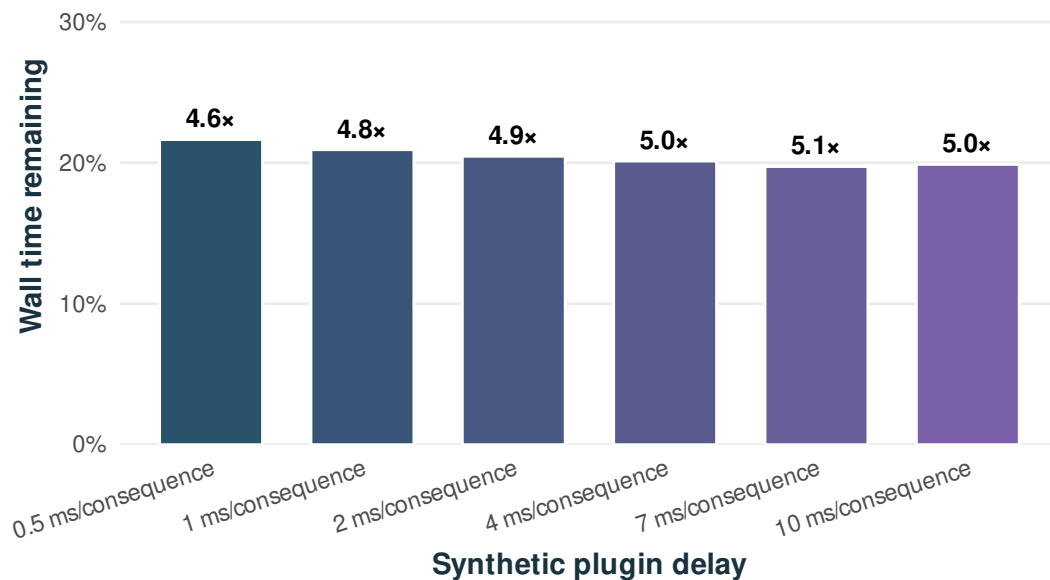


VCFcache converts annotation complexity into time saved

One 4.80-million-variant real WGS · bundled Zenodo gnomAD AF $\geq 10\%$ cache · ~80% hits · light statistics · no technical repeats

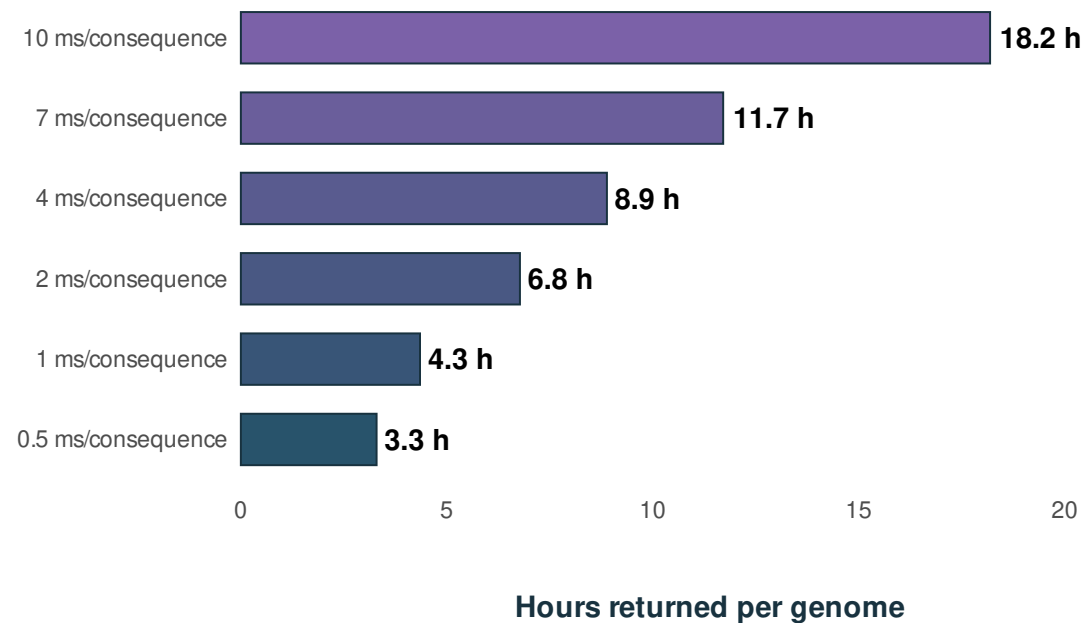
A About 80% reuse removes most of every pipeline

Numbers above bars are measured speedups



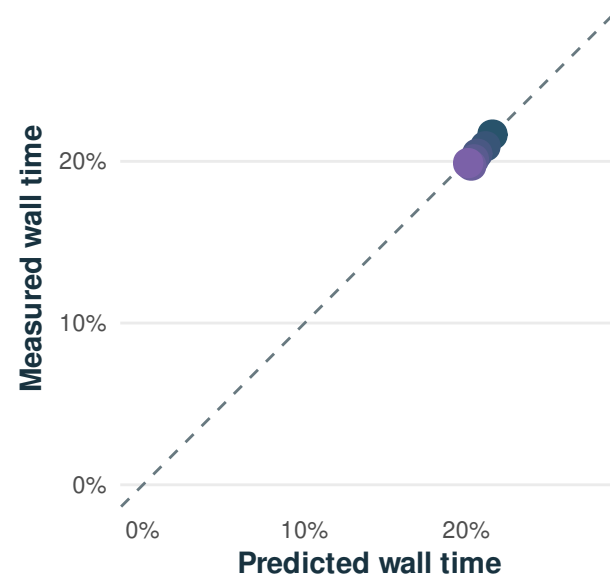
B Slower pipelines return hours, not minutes

The same real WGS and cache for every load



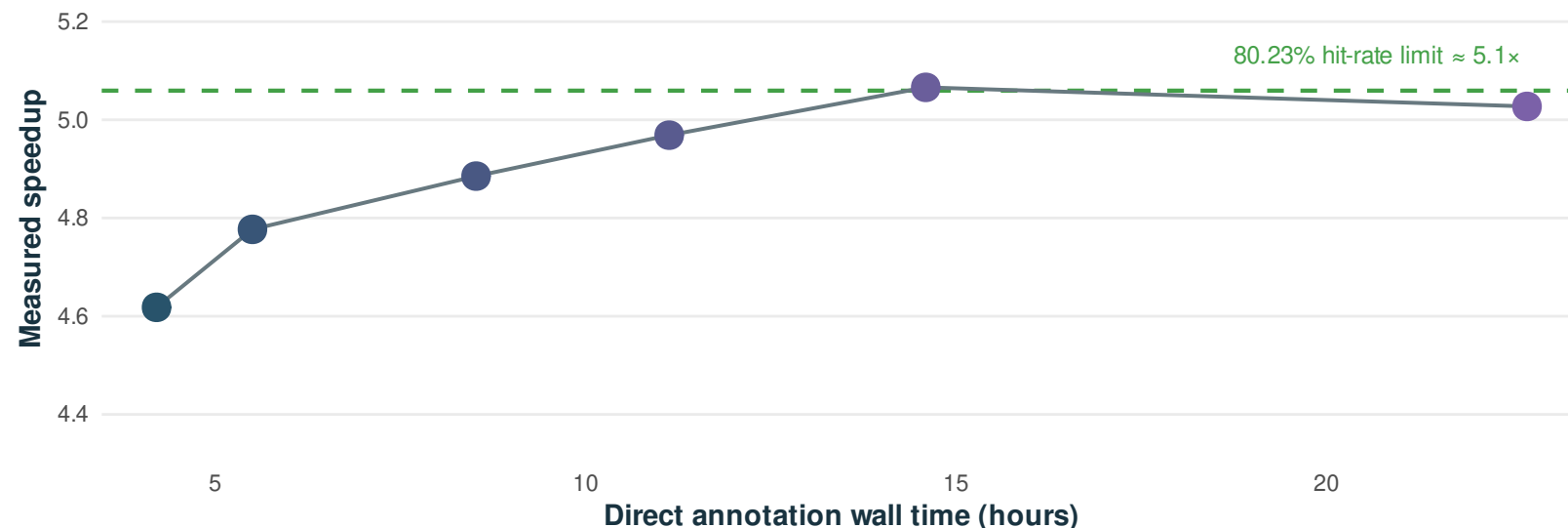
C Simple runtime model stays accurate

VEP lookup overhead + miss fraction $\times$ direct runtime



D Speedup reaches its hit-rate ceiling

Absolute time returned keeps growing as pipelines slow



All six cached outputs were semantically equivalent to their pipeline-specific uncached output. Synthetic delays run only for transcript consequences submitted to VEP and emit no annotation fields.